



G.T E C H

PROJECT SPECIFICATIONS

Capstone Project - GTech 4

PROJECT

FORMAT

Individual project

PROJECT CONTEXT

As part of the MSc in Video Game Development, Real-Time 3D Rendering and Immersive Technologies, you will progressively design and develop a complete sandbox game in Unreal Engine.

The project will be carried out over 9 weeks, divided into 3 blocks of 3 weeks.

Rather than a succession of independent projects, the different weeks represent development stages of the same project. Each week introduces new systems, improves existing elements, and progressively evolves the prototype into a complete experience.

The objective is to place you in a workflow close to real production: start from a simple technical foundation, progressively build the various game systems, make them communicate with one another, improve them, and then integrate them into a coherent experience.

PROJECT DESCRIPTION

You must design a sandbox game in which the player can explore an environment, interact with different systems, and experiment with mechanics based in particular on physics and the capabilities offered by Unreal Engine.

The concept, setting, and direction of the project are open.

However, your game must be built around one essential principle:

The player must be able to experiment with the game systems and trigger reactions in the environment.

The project should therefore prioritize systemic interactions, physics, environmental reactions, and opportunities for experimentation rather than heavily scripted progression or gameplay based solely on a succession of objectives.

The gameplay may remain relatively simple. Priority will be given to the quality of the systems developed, their interactions, and the level of technical mastery demonstrated.

OVERALL OBJECTIVE

The student must be able to design, develop, test, optimize, and present technical prototypes in Unreal Engine.

The work produced must demonstrate progressive mastery of:

- Blueprint/C++ architecture;
- real-time physics and Chaos;
- interactive system design;
- rendering architecture and pipeline;
- materials, shaders, Niagara, and VFX;
- CPU/GPU optimization;
- interactive staging;
- camera, lighting, and sound;
- environmental storytelling and player experience;
- debugging, profiling, and technical documentation.

These skills correspond to the overall objectives defined for all three blocks.

PROJECT ORGANIZATION

BLOCK 1 — Gameplay, C++/Blueprint & Advanced Physics

- **Purpose: develop mastery of Unreal Engine gameplay and physics systems.**
- The project must progressively incorporate interaction systems, physics mechanics, Chaos, and a coherent, modular Blueprint/C++ architecture.
- The block will culminate in a physics-based gameplay prototype demonstrating technical mastery of the systems developed.

BLOCK 2 — Real-Time Rendering

- **Purpose: understand and leverage Unreal Engine's graphics pipeline.**
- The project must provide opportunities to experiment with and then integrate materials, shaders, VFX, Niagara, lighting, post-processing, and advanced graphics systems, with particular attention to profiling and GPU optimization.
- The block will culminate in a technically controlled interactive visual experience.

BLOCK 3 — Staging & Player Experience

- **Purpose: use Unreal Engine as a tool for directing and shaping the player experience.**
- The project must integrate staging, camera work, Sequencer, lighting, sound design, interactive events, pacing, and environmental storytelling.
- The block will culminate in a coherent, immersive, and technically stable prototype.

GENERAL REQUIREMENTS

All work produced must comply with the following requirements:

- Individual work.
- Development must be carried out in Unreal Engine.
- Consistent use of Blueprint and/or C++ according to the needs of the project.
- Clear, modular, and maintainable architecture.
- Functional and testable prototype.
- Scope appropriate to the available production time.
- The student must understand and master the systems they implement.
- Technical choices must be explainable and justifiable.
- The project must be regularly tested, debugged, and, where relevant, profiled.
- External assets must comply with their terms of use.
- Technical quality and mastery of the systems take priority over the quantity of content.

Each week is individual; weeks 1 and 2 of each block result in a graded submission, while week 3 results in a submission

accompanied by an oral presentation. In addition, two projects during the year must be completed entirely in English.

EXPECTED DELIVERABLES

Depending on the week and block concerned, the submission may include:

- the complete Unreal project;
- a playable version/build when requested;
- the Blueprint/C++ source files;
- the assets required for the project to function;
- brief technical documentation;
- materials that make it possible to understand the architecture and systems developed;
- any profiling/debugging results;
- a presentation or oral defense at the end of each block.

The specific requirements will be defined in the specifications for each week.

TECHNICAL REQUIREMENTS

A successful project will not be assessed solely on its visual quality or gameplay outcome.

The student must demonstrate:

Functionality — The required systems are operational.

Architecture — The project is properly structured and sufficiently modular.

Technical mastery — The student understands and can explain what they have developed.

Robustness — The prototype is stable and the main problematic cases have been addressed.

Performance — Technical choices remain consistent with real-time constraints.

Readability — The project and its architecture can be understood and assessed by a third party.

Iteration — The student is able to identify a problem, test a solution, and improve the prototype.

This approach reflects the program's expectations regarding professional practice, autonomy, debugging, scope management, and documentation.

USE OF ARTIFICIAL INTELLIGENCE

The use of generative AI tools is permitted as an assistance tool, particularly for research, ideation, documentation, debugging, comparing approaches, or accelerating certain tasks.

However, it must never replace the student's technical understanding.

Any element created with the help of AI must be understood, verified, tested, adapted to the project, and explainable by the student.

When AI has been used for a submission, an AI statement of intent is mandatory. It must specify, in particular, where, why, and how AI was used, and justify its value as well as its limitations. The minimum length is one page for projects and two pages for the midterm assessment.

ASSESSMENT CRITERIA

The assessment of the different submissions will take into account, in particular:

Technical

- quality of the architecture;
- mastery of Unreal systems;
- quality of the implementations;
- stability and performance.

Design

- relevance of technical choices;
- coherence of the systems;
- ability to stay within the required scope.

Experience

- clarity of interactions;
- quality of feedback;
- coherence of the proposed experience.

Professional Practice

- autonomy;
- organization;
- debugging and analytical ability;
- documentation;
- ability to present and defend the choices made.

AI

- transparency of use;
- relevance of its use;
- critical perspective;
- genuine mastery of the elements produced with its assistance.

These principles are aligned with the criteria used for the final assessment, which distinguishes architecture/code, physics,

rendering, user experience, staging, and cross-cutting AI/documentation assessment.

EXPECTED FINAL OUTCOME

At the end of the 9 weeks, the student must be able to independently produce a technically controlled interactive experience in Unreal Engine combining:

Gameplay / Physics + Real-Time Rendering + Staging / Player Experience

The final objective is not to produce a large-scale game, but to demonstrate a solid understanding of the engine, clean architecture, rapid prototyping ability, and professional mastery of the systems developed.



EDUCATIONAL AND PROFESSIONAL OBJECTIVES OF THE PROJECT

Skills Developed

- Develop advanced gameplay systems in Unreal Engine
- Structure a coherent and maintainable Blueprint/C++ architecture
- Design real-time physics interactions
- Use Chaos for gameplay and simulation purposes
- Develop advanced physics-based gameplay mechanics
- Design modular interactive systems
- Understand Unreal Engine gameplay architecture
- Debug and profile an Unreal prototype
- Produce technical prototypes under time constraints
- Understand the impact of physics systems on gameplay and game feel
- Understand Unreal Engine's real-time rendering pipeline
- Develop advanced rendering systems
- Create complex materials and shaders
- Use Niagara for technical purposes
- Develop interactive VFX
- Understand CPU/GPU interactions in a real-time engine
- Use compute shaders in Unreal Engine
- Integrate advanced lighting systems
- Optimize graphics and GPU performance
- Produce technically coherent visual experiences
- Build an immersive player experience
- Use Unreal Engine's staging tools
- Develop interactive and cinematic sequences
- Master camera systems and player direction
- Design environmental storytelling
- Structure the pacing of an interactive experience
- Use lighting and sound as directing tools
- Produce a clear and coherent user experience
- Organize a real-time production pipeline
- Present an Unreal Engine prototype from a technical perspective

Soft Skills Developed

- Technical autonomy
- Analytical and problem-solving skills
- Time and production scope management
- Technical rigor
- Ability to iterate quickly on a prototype
- Organization of a personal workflow
- Adaptability to technical constraints
- Critical thinking regarding technical choices
- Ability to document a technical project
- Clear technical communication
- Ability to structure a prototyping approach
- Production priority management
- Technology monitoring
- Ability to receive and incorporate feedback
- Awareness of clarity and user experience
- Understanding of performance and optimization challenges
- Methodical approach to debugging
- Awareness of the overall coherence of a project
- Responsible and transparent use of artificial intelligence tools
- Professionalization of Unreal Engine development practices



INSTRUCTOR

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BIOGRAPHY

PhD in Computer Science (CNAM Paris), specializing in difficulty in video games and its relationship with player motivation. A researcher and developer of a dynamic difficulty adjustment model, he is the author of several scientific publications on the subject.

For the past six years, he has worked as a freelance instructor in video games and artificial intelligence and has taught at Gaming Campus since 2022, combining scientific expertise with hands-on teaching to train and inspire the next generation of industry professionals.